

Ex. No. 5	JAVASCRIPT - DOM ELEMENTS AND EVENTS
Date of Exercise	21.08.2026
GitHub Host Link	https://github.com/jenohsam2006/WT_LAB.git

Aim

To create web pages which demonstrate the applications of document object model and events in JavaScript.

Description

Document Object Model:

A Document object represents the HTML document that is displayed in that window. The Document object has various properties that refer to other objects which allow access to and modification of document content. The way a document content is accessed and modified is called the Document Object Model, or DOM.

Finding HTML Element by Id

```
var myElement = document.getElementById("intro");
```

Finding HTML Elements by Tag Name

```
var x = document.getElementsByTagName("p");
```

Finding HTML Elements by Class Name

```
var x = document.getElementsByClassName("intro");
```

Finding HTML Elements by CSS Selectors

```
var x = document.querySelectorAll("p.intro");
```

Attach event handler to the document

```
document.addEventListener("click", function()  
{  
    document.getElementById("demo").innerHTML = "Karunya";  
});
```

JavaScript Events:

JavaScript's interaction with HTML is handled through events that occur when the user or the browser manipulates a page.

Event	Description
onabort	Triggers on an abort event.
onblur	Triggers when the window loses focus.
onchange	Triggers when an element changes.
onclick	Triggers on a mouse click.
ondblclick	Triggers on a mouse double-click.
ondrag	Triggers when an element is dragged.
onfocus	Triggers when the window gets focus.
onkeyup	Triggers when a key is released.
onload	Triggers when the document loads.
onkeydown	Triggers when a key is pressed.
onkeypress	Triggers when a key is pressed and released.
onmousedown	Triggers when a mouse button is pressed.
onmousemove	Triggers when the mouse pointer moves.
onmouseout	Triggers when the mouse pointer moves out of an element.
onmouseover	Triggers when the mouse pointer moves over an element.
onmouseup	Triggers when a mouse button is released.
onresize	Triggers when the window is resized.
onscroll	Triggers when an element's scrollbar is being scrolled.
onsubmit	Triggers when a form is submitted.
onunload	Triggers when the user leaves the document.

Sample Questions:

1. Write a function that accepts a string as a parameter and find the shortest and longest word within the string. Example string: 'Web Development Tutorial'

Expected Output: Shortest: 'Web'

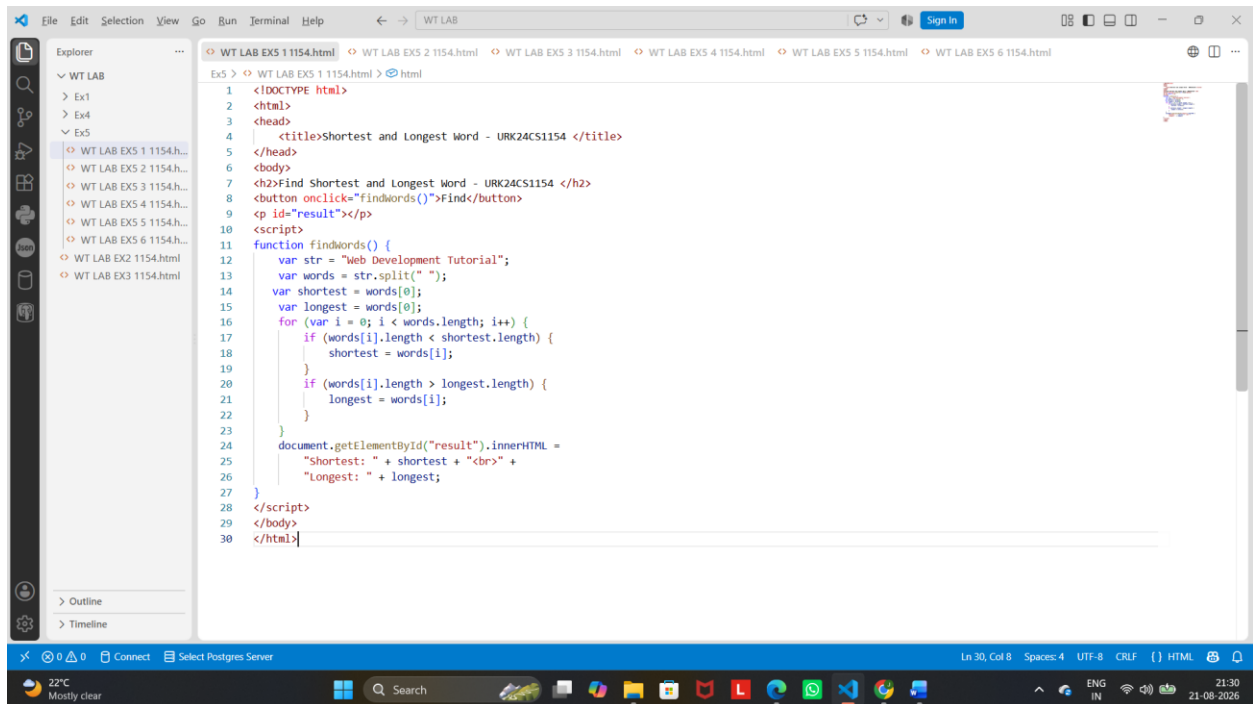
Longest: 'Development'

Hint: Split a string into an array - split()

Find the length of a string - stringname.length

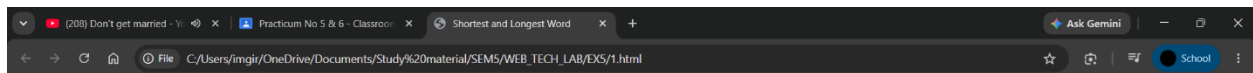
Find the size of an array - arrayname.length

Program



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Shortest and Longest Word - URK24CS1154 </title>
5 </head>
6 <body>
7   <h2>Find Shortest and Longest Word - URK24CS1154 </h2>
8   <button onclick="findwords()">Find</button>
9   <p id="result"></p>
10  <script>
11    function findwords() {
12      var str = "Web Development Tutorial";
13      var words = str.split(" ");
14      var shortest = words[0];
15      var longest = words[0];
16      for (var i = 0; i < words.length; i++) {
17        if (words[i].length < shortest.length) {
18          shortest = words[i];
19        }
20        if (words[i].length > longest.length) {
21          longest = words[i];
22        }
23      }
24      document.getElementById("result").innerHTML =
25        "Shortest: " + shortest + "<br>" +
26        "Longest: " + longest;
27    }
28  </script>
29 </body>
30 </html>
```

Output



Find Shortest and Longest Word

hiiii long

Shortest: long

Longest: hiiii

2. BMI Calculator:

The following HTML snippet shows the skeleton of a BMI calculator. Write the calculate() function that takes the weight and height in the text boxes and displays the BMI in the span with the id of “score”. BMI is calculated from the equation: $\text{weight} / (\text{height} * \text{height})$. You should also describe the person's body type based on their BMI score using the following criteria:

- Underweight: < 18
- Normal: 18 - 25
- Overweight: 25-30
- Obese: > 30

The person's body type should be displayed in the span with the id of "type" and you may assume valid input into the text boxes.

```
<h1>Calculate your Body Mass Index (BMI) !</h1>
<fieldset>
Weight (in kilograms): <input type="text" id="weight" /><br />
Height (in meters): <input type="text" id="height" /><br />
<button onclick="calculate();">Get BMI !< /button>
</fieldset>
<p>
Your BMI is <span id="score"></span>. <br />
Your body type is <span id="type"></span>.
</p>
```

```
function calculate () {
//extract data from input box
    var height = parseFloat(document.getElementById("height").value);
//change the content of html element
    document.getElementById("score").innerHTML = bmi;
}
```

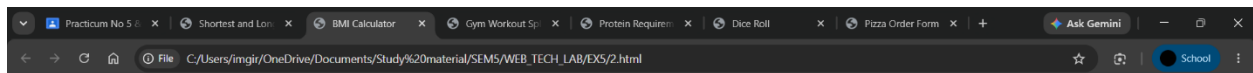
Program

```

1  <DOCTYPE HTML>
2  <html>
3  <head>
4    <title>BMI Calculator</title>
5  </head>
6  <body>
7
8    <h3>Calculate your Body Mass Index (BMI)</h3>
9
10   <fieldset>
11
12     Weight (in kilograms):
13     <input type="text" id="weight"><br><br>
14
15     Height (in meters):
16     <input type="text" id="height"><br><br>
17
18     <button onclick="calculate()">Get BMI</button>
19
20   </fieldset>
21
22   <p>
23     Your BMI is <span id="score"></span><br>
24     Your body type is <span id="type"></span>
25   </p>
26
27   <script>
28   function calculate() {
29
30     // extract data from input box
31     var weight = parseFloat(document.getElementById("weight").value);
32     var height = parseFloat(document.getElementById("height").value);
33
34     // calculate BMI
35     var bmi = weight / (height * height);
36
37     // display BMI
38     document.getElementById("score").innerHTML = bmi.toFixed(2);
39
40     // determine body type
41     if (bmi < 18) {
42       document.getElementById("type").innerHTML = "Underweight";
43     }
44     else if (bmi >= 18 && bmi < 25) {
45       document.getElementById("type").innerHTML = "Normal";
46     }
47     else if (bmi >= 25 && bmi <= 30) {
48       document.getElementById("type").innerHTML = "Overweight";
49     }
50     else {
51       document.getElementById("type").innerHTML = "Obese";
52     }
53   }
54   </script>

```

Output



Calculate your Body Mass Index (BMI)!

Weight (in kilograms):

Height (in meters):

Your BMI is 0.00.
Your body type is Underweight.



3. Fruit Farm:

The following HTML snippet is from a webpage that allows users to design rectangular fruit farm beds. When you click the "Plant Garden" button, the page should generate a rectangular grid of fruit images in the div with the id of "garden." The dimensions of grid are given by the values entered in the text boxes. Write the grow() JavaScript function to complete the functionality for the webpage. You should randomly display either "apple.jpg" or "orange.jpg" for each fruit image, and either fruit should show up with equal probability.

```
<h1>Fruit Farm</h1>
<div>
  Rows: <input type="text" id="rows" />
  Columns: <input type="text" id="columns" /> <br />
  <button onclick="grow();">Plant Garden</button>
</div>
<div id="garden"></div>
```

```
//add image
```

```
document.getElementById("garden").innerHTML += "<img src= Apple.jpg width =100px
height=100px>"; // += for appending the text
```

```
// create an array of strings
```

```
var arr = ["Orange.jpg","Apple.jpg"];
```

```
//generate random number in the range of 0 to 10
```

```
var number = Math.floor((Math.random() *upper bound) +lower bound);
```

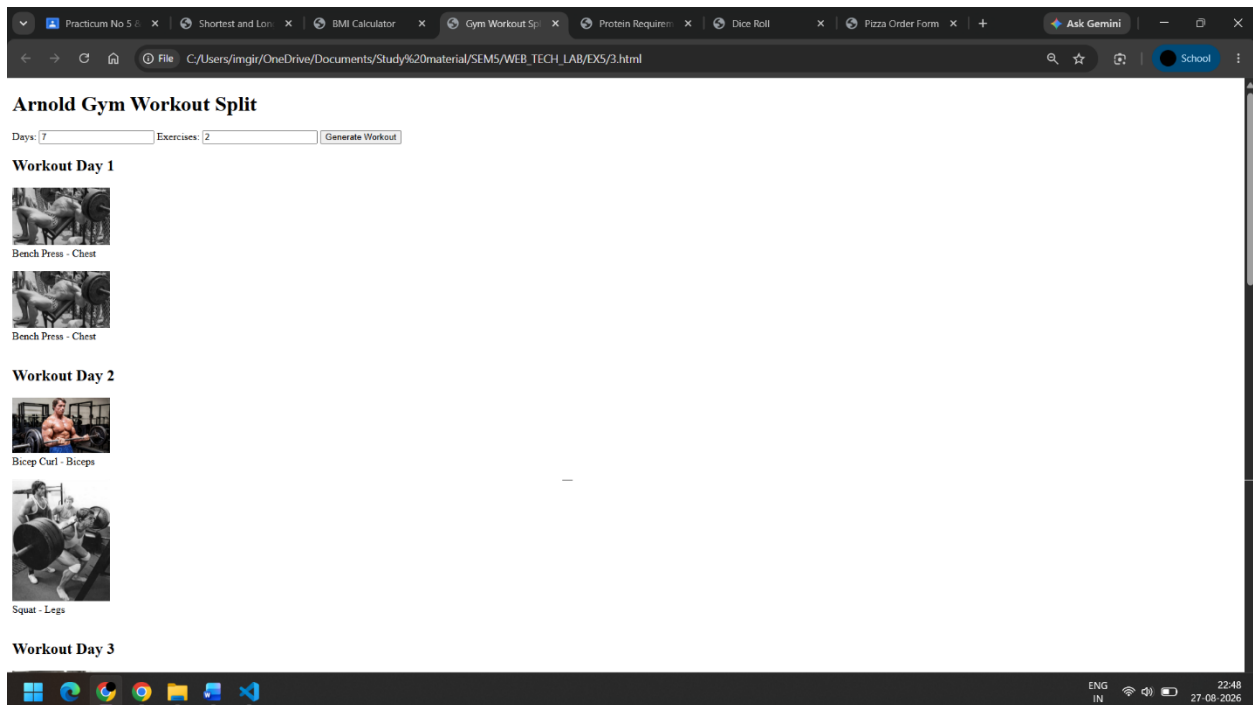
Program

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>Gym Workout Split</title>
5  </head>
6  <body>
7
8    <h1>Arnold Gym Workout Split</h1>
9
10   Days:
11   <input id="days" type="number">
12
13   Exercises:
14   <input id="ex" type="number">
15
16   <button onclick="generate()">Generate Workout</button>
17
18   <div id="workout"></div>
19
20   <script>
21
22   function generate() {
23
24     var list = [
25       ["bench.jpg", "Bench Press", "Chest"],
26       ["biceps.jpg", "Bicep Curl", "Biceps"],
27       ["squat.jpg", "Squat", "Legs"],
28       ["shoulder.jpg", "Arnold Press", "Shoulders"],
29       ["back.jpg", "Lat Pulldown", "Back"],
30       ["triceps.jpg", "Tricep Extension", "Triceps"]
31     ];
32
33     var d = document.getElementById("days").value;
34     var e = document.getElementById("ex").value;
35     var out = "";
36
37     for (var i = 1; i <= d; i++) {
38
39       out += "<h2>Workout Day " + i + "</h2>";
40
41       for (var j = 0; j < e; j++) {
42
43         var x = list[Math.floor(Math.random() * list.length)];
44
45         out += "<div>" +
46           "<img src='" + x[0] + "' width='150'><br>" +
47           x[1] + " - " + x[2] +
48           "</div><br>";
49       }
50     }
51
52     document.getElementById("workout").innerHTML = out;
53   }
54

```

Output



4. The Currency Converter:

The following HTML snippet shows the skeleton of a rupees-to-dollars, dollars-to-rupees conversion tool. Write the `convert()` function that takes the value in the text input and converts it from the unit selected in the left dropdown box to the unit selected on the right. The unit should be displayed in the empty span with the id of “answer.” The conversion factor from rupees to dollars is 0.015, and the conversion factor from dollars to rupees is 67.12. You should edit the HTML to add ids to the elements as necessary and you may assume valid input.

```
<fieldset> Convert <input type="text" id="data" />
<select id="from">
  <option>Indian Rupee</option>
  <option>US Dollar</option>
</select>
to
<select id="to">
  <option> US Dollar </option>
  <option> Indian Rupee </option>
</select>
<button
  onclick="convert();">Calculate</button>
  id="answer"></span>
</fieldset>
```

//extract value from drop down list

```
var index = document.getElementById("from").selectedIndex; // return index
```

```
var value = document.getElementById("from")[index].value; // return value
```

Program

```

1 <DOCTYPE html>
2 <html>
3 <head>
4 <title>Protein Requirement Calculator</title>
5 </head>
6 <body>
7 <h1>Protein Requirement Calculator</h1>
8 </body>
9 </html>
10
11 <script>
12
13   Weight (in kg)
14   <input type="text" id="weight">
15   <br>
16   Activity / Goal:
17
18   <select id="goal">
19     <option value="0.8">Low Activity - 0.8 g/kg</option>
20     <option value="1.2">Moderately Active - 1.2 g/kg</option>
21     <option value="1.6">Muscle Building - 1.6 g/kg</option>
22     <option value="1.8">Very Active - 1.8 g/kg</option>
23   </select>
24
25   <br>
26   <button onclick="calculateProtein()">Calculate Protein</button>
27
28 </script>
29
30 <div>
31   Recommended Protein:
32   <span id="recommendation"></span>
33 </div>
34
35 <div>
36   Daily Protein Requirement:
37   <span id="answer"></span> grams
38 </div>
39
40 </script>
41
42 function calculateProtein() {
43   // Get weight from input box
44   var weight = parseInt(document.getElementById("weight").value);
45   // Get selected protein requirement
46   var protein = parseInt(document.getElementById("goal").value);
47   // Calculate daily protein
48   var result = weight * protein;
49   // Display recommendation
50   document.getElementById("recommendation").innerHTML =
51     protein + " grams per kg";
52   // Display total protein
53   document.getElementById("answer").innerHTML =
54     "Result: " + result;
55 }
56
57 </script>
58
59 </body>
60 </html>

```

Output

Weight (in kg):

Activity / Goal: Low Activity - 0.8 g/kg

Result

Recommended Protein:

Daily Protein Requirement: grams

Show desktop

5. Coin Flip:

Write JavaScript code to randomly change the src of the img in the HTML snippet to "heads.jpg" or "tails.jpg" when the "Flip!" button is pressed. The image should change to the heads or tails picture with equal probability. Keep track of how many flips have resulted in heads and tails and report the results beneath the button in two spans with ids "num_heads" and "num_tails" respectively. You should write JavaScript code to attach the event handler to the button and you may assume the user enters valid input.

```
<h1>Heads or Tails?</h1>
<div>
   <br /> <button id="flip">Flip!</button>
</div>
<p>
  Heads Total: <span id="num_heads">0</span><br />
  Tails Total: <span id="num_tails">0</span>
</p>
```

```
// attach the event handler to the button
```

```
document.getElementById("flip").addEventListener("click", flipFunction);
```

```
function flipFunction() {
```

```
    //code goes here
```

```
}
```

```
// change the src of the img in html
```

```
    document.getElementById("coin").src = "head.jpg";
```

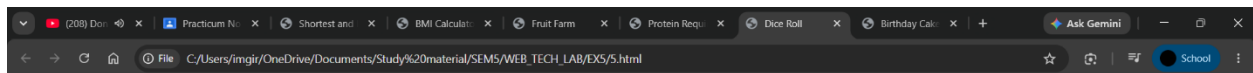
Program

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Dice Roll</title>
5 </head>
6 <body>
7
8 <div id="dice">
9
10 <div>
11
12 <div id="dice" style="font-size: 30px; text-align: center; width: 100px; height: 100px; margin: 0 auto; border: 1px solid black; display: flex; align-items: center; justify-content: center;">
13
14 <div id="roll" style="border: 1px solid black; padding: 5px; width: 100px; height: 40px; margin: 0 auto; text-align: center;">Roll Dice</div>
15
16 </div>
17
18 </div>
19
20 <div id="result"></div>
21
22 </div>
23
24 <div id="total"></div>
25
26 </div>
27
28 <script>
29
30 var totalRolls = 0;
31
32 // Attach event handler to the button
33 document.getElementById("roll").addEventListener("click", rollDice);
34
35 function rollDice() {
36
37 // Generate random number from 1 to 6
38 var number = Math.floor(Math.random() * 6) + 1;
39
40 // Dice symbols
41 var diceFaces = ["1", "2", "3", "4", "5", "6"];
42
43 // Change dice face
44 document.getElementById("dice").innerHTML =
45 diceFaces[number - 1];
46
47 // Display result
48 document.getElementById("result").innerHTML = number;
49
50 // Increase total rolls
51 totalRolls++;
52
53 document.getElementById("total").innerHTML = totalRolls;
54
55 }
56
57 </script>
58
59 </body>
60 </html>

```

Output



Roll the Dice!



Roll Dice

Last Roll: 5

Total Rolls: 6



6. Birthday Cake Order Form:

The following HTML snippet represents a skeleton of a price calculator for a cake order form. A 1/2kg cake is Rs.100, a 1kg cake is Rs.200 and a 2kg cake is Rs.400. Tax is an additional 9% to the order. The user must also pay a tip from 10-20% of the total cost of the order after tax. Write the JavaScript code necessary to calculate and display the total cost of the order in the span with the id of "price" when the user clicks the "Calculate Order" button. You should write unobtrusive JavaScript code to attach the event handler to the button and you may assume the user enters valid input.

```
<h1>Pizza Order Form</h1>
<fieldset>
# of 1/2kg Cakes <input type="text" id="small" /><br />
# of 1kg Cakes <input type="text" id="medium" /><br />
# of 2kg Cakes <input type="text" id="large" /><br /> Tip:
<label><input type="radio" value="10" name="tip" /> 10%</label>
<label><input type="radio" value="15" name="tip" checked="checked" /> 15% </label>
<label><input type="radio" value="20" name="tip" /> 20%</label> <br />
<button id="submit">Calculate Order</button> </fieldset>
<p>
Cost of order: <span id="price"> </span>
</p>
```

//extract value from radio buttons

```
document.querySelector('input[name="tip"] : checked').value
```

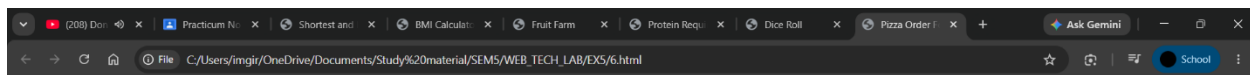
Program

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Birthday Cake Order Form - URK24CS1154 </title>
5 </head>
6 <body>
7   <h2>Birthday Cake Order Form - URK24CS1154 </h2>
8   Number of 1/2kg Cakes:
9   <input type="text" id="small"><br><br>
10  Number of 1kg Cakes:
11  <input type="text" id="medium"><br><br>
12  Number of 2kg Cakes:
13  <input type="text" id="large"><br><br>
14  Tip:
15  <input type="radio" name="tip" value="10"> 10%
16  <input type="radio" name="tip" value="15" checked=""> 15%
17  <input type="radio" name="tip" value="20"> 20%
18  <br><br>
19  <button id="submit">Calculate Order</button>
20  <p>Cost of order: Rs. <span id="price"></span></p>
21  <script>
22    document.getElementById("submit").addEventListener("click", calculate);
23    function calculate() {
24      let small = parseInt(document.getElementById("small").value);
25      let medium = parseInt(document.getElementById("medium").value);
26      let large = parseInt(document.getElementById("large").value);
27      let tip = parseInt(
28        document.querySelector('input[name="tip"]:checked').value
29      );
30      let cost = (small * 100) +
31        (medium * 200) +
32        (large * 400);
33      let tax = cost * 0.09;
34      let total = cost + tax;
35      total = total + (total * tip / 100);
36      document.getElementById("price").innerHTML =
37        total.toFixed(2);
38    }
39  </script>
40 </body>
41 </html>

```

Output



Pizza Order Form

Small Pizza:

Medium Pizza:

Large Pizza:

Delivery: ☐ Pickup ☒ Home Delivery

Cost of Order: Rs. 1750.00

Result

Thus, the JavaScript DOM Elements and Events programs were successfully executed. The programs demonstrated DOM manipulation, event handling, input processing, random values, calculations, and dynamic changes to HTML elements.